

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import math
from collections import Counter
```

```
In [2]: data = pd.read_csv("KNN.csv")
data
```

Out[2]:

	OBS	X1	X2	X3	Y
0	1	0	3	0	RED
1	2	2	0	0	RED
2	3	0	1	3	RED
3	4	0	1	2	GREEN
4	5	-1	0	1	GREEN
5	6	1	1	1	RED

```
In [3]: # X_Training Data
X_Train = data.iloc[:, 1:-1].values
X_Train
```

```
Out[3]: array([[ 0,  3,  0],
               [ 2,  0,  0],
               [ 0,  1,  3],
               [ 0,  1,  2],
               [-1,  0,  1],
               [ 1,  1,  1]], dtype=int64)
```

```
In [4]: # Y_Train
Y_Train = data.iloc[:, -1].values
Y_Train
```

```
Out[4]: array(['RED', 'RED', 'RED', 'GREEN', 'GREEN', 'RED'], dtype=object)
```

```
In [5]: # Defining Euclidean Distance

def Euclidean_Distance(p1,p2):
    return math.sqrt(sum((a-b) ** 2 for a,b in zip(p1,p2)))
```

In [6]: *# Defining KNN Algorithm*

```
def KNN_Classify(X_Train, Y_Train, Test_Point, K):
    distances = []

    #Calculating Euclidean Distance
    for i in range(len(X_Train)):
        dist = Euclidean_Distance(Test_Point, X_Train[i])
        distances.append((dist, Y_Train[i]))

    print(distances)

    # Step 4: Sort distances and pick k nearest
    distances.sort(key=lambda x : x[0])
    k_nearest = [label for _, label in distances[:k]]

    # Step 5: Most Occured
    most_common = Counter(k_nearest).most_common(1)
    return most_common[0][0]
```

In [19]: *# Example test point*

```
Test_Point = [0, 0, 0] # Test Point

k = 3 # K value

predicted_label = KNN_Classify(X_Train, Y_Train, Test_Point, k) # Prdiction

[(3.0, 'RED'), (2.0, 'RED'), (3.1622776601683795, 'RED'), (2.23606797749979, 'GREEN'), (1.4142135623730951, 'GREEN'), (1.7320508075688772, 'RED')]
```

In [20]: *# Output*

```
print(f"Test Point is : {Test_Point}")
print(f"The Predicted Value is: {predicted_label}")

Test Point is : [0, 0, 0]
The Predicted Value is: RED
```

In []: